

Sample-Efficient Reinforcement Learning for Robotics

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ABSTRACT:

We present a sample-efficient RL algorithm for robotic manipulation that combines model-based planning with offline pre-training.

Our approach reduces sample complexity by 10x compared to standard model-free methods on robotic grasping tasks.

1. PROBLEM FORMULATION

Robotic manipulation requires learning complex motor skills with limited real-world interaction due to safety and cost constraints.

2. PROPOSED METHOD: MOPO-Robot

Model-based Offline Policy Optimization for Robotics

2.1 Offline Pre-training Phase

- Collect diverse manipulation dataset (100K trajectories)
- Train world model using Transformer architecture
- Learn value function via conservative Q-learning

2.2 Online Fine-tuning Phase

- Use world model for planning and exploration
- Update policy with real robot feedback
- Uncertainty-aware model predictions

2.3 Safety Constraints

Implement collision avoidance and joint limit constraints.

3. EXPERIMENTAL EVALUATION

Tasks: Block stacking, peg insertion, door opening

Robot platform: Franka Emika Panda 7-DOF arm

Simulation: PyBullet and MuJoCo environments

Results:

- Block stacking: 95% success rate (vs 60% baseline)
- Peg insertion: 87% success rate (vs 45% baseline)
- Sample efficiency: 10x improvement over SAC

4. ABLATION STUDIES

Key components contributing to performance:

- Offline pre-training: +40% success rate
- Model-based planning: +25% success rate
- Uncertainty estimation: +15% success rate